

Sitong Zhang

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Research Interest

Reinforcement Learning (RL); Distributed Machine Learning (ML) systems

Work Experience

Aalto University, Department of Computer Science, Postdoctoral Researcher

Espoo, Finland
2025.5—present

Supervisor: Prof. Bo Zhao

Data-Centric Runtime Systems for Distributed RL

- Design a runtime system that optimizes data movement, placement, and access along the RL training pipeline to improve scalability and training efficiency.
- Tech Stack: Python, RLLib, profiling tools (Nsys, torch profiler)

Elastic Runtime Systems for Large-Scale RL Post-Training

- Develop a adaptive runtime system that dynamically adapts resource allocation and execution configurations for LLM RL post-training under evolving workload and resource conditions.
- Tech Stack: Python, frameworks (Ray, Megatron, DeepSpeed, vLLM, Verl), torch profiler

Centre for Intelligent Multidimensional Data Analysis Limited, Postdoctoral Fellow

Hong Kong
2024.2—2024.7

Supervisor: Prof. Hong Yan

AI-Driven Intelligent Indoor Positioning on Edge Devices

- Proposed a multimodal approach to enhance the accuracy of the indoor positioning system by fusing WiFi RSSI and IMU sensor data.
- Developed an AI-driven Android application that integrates the proposed method, achieving real-time multi-floor user location inference on edge devices, such as mobile phones.
- Tech Stack: Python, Java, Android Studio

Education

Ph.D. Harbin Engineering University, Information and Communication Engineering 2018.9 — 2023.12

Supervisor: Prof. Yibing Li

Research Focus: Deep Reinforcement Learning (DRL), Unmanned Aerial Vehicle (UAV), Motion Planning.

Tech Stack: Python, C#, C++, Unity 3D, Gazebo, Robot Operating System (ROS).

B.E. Harbin Engineering University, Information and Communication Engineering 2014.9 — 2018.6

Selected Publications

- **Sitong Zhang**, Yibing Li, Qianhui Dong. *Autonomous navigation of UAV in multi-obstacle environments based on a Deep Reinforcement Learning approach*. Applied Soft Computing, 2022.
- **Sitong Zhang**, Yibing Li, Fang Ye, Xiaoyu Geng, Zitao Zhou, Tuo Shi. *A Hybrid Human-in-the-Loop Deep Reinforcement Learning Method for UAV Motion Planning for Long Trajectories with Unpredictable Obstacles*. Drones, 2023.
- **Sitong Zhang**, Yibing Li, Qian Sun, Fang Ye. *QoS maximization scheduling of multiple UAV base stations in 3D environment*. Internet of Things, 2023.
- **Sitong Zhang**, Yibing Li, Yuan Tian, Zitao Zhou, Xiaoyu Geng, Tuo Shi. *Dynamic Redeployment of UAV Base Stations in Large-Scale and Unreliable Environments*. Internet of Things, 2023.
- Yibing Li, **Sitong Zhang**, Fang Ye, Tao Jiang, Yingsong Li. *A UAV path planning method based on deep reinforcement learning*. 2020 IEEE USNC-CNC-URSI North American Radio Science Meeting (Joint with AP-S Symposium). IEEE, 2020.

Academic Service

Technical Reviewer

- Applied Soft Computing
- Journal of Intelligent & Robotic Systems
- Discover Computing
- International Journal of Machine Learning and Cybernetics
- Artificial Intelligence Review
- Scientific Reports

Additional Experience And Awards

Outstanding Graduate, Harbin Engineering University, 2018

Honorable Mention in Interdisciplinary Contest in Modeling (ICM), 2016

Second Prize in China Undergraduate Mathematical Contest in Modeling, 2015

The 12th International Collegiate Snow Sculpture Contest, 2020.1

Volunteer with the University of Technology Sydney Team

The International Marine Vehicle Design and Construction Invitational Contest, 2019.8

Volunteer with the Cardiff University Team

Technologies

Programming Languages: Python, C#, C++, Java

Distributed ML Systems: frameworks (e.g., DeepSpeed, Megatron, Verl, Ray), profiling tools (e.g., Nsight system (Nsys), torch profiler), HPC environments (slurm-based clusters)

Simulation & Robotics: Unity 3D, Gazebo, Robot Operating System (ROS)